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Coastal fish assemblages reflect marine habitat connectivity and ontogenetic shifts in an estuary-bay-continental shelf gradient

José Amorim Reis-Filho[a,b,*], Kurt Schmid[b], Euan Sinclair Harvey[c], Tommaso Giarrizzo[b]

[a] ICHTUS soluções em meio ambiente, sl 306, Rua Macaubas, Rio Vermelho, Salvador, BA, 41940-250, Brazil
[b] Núcleo de Ecologia Aquática e Pesca da Amazônia, Universidade Federal do Pará, Av. Perimetral 2561, Terra Firme, Belém, PA, 66040-170, Brazil
[c] School of Molecular and Life Sciences, Curtin University, Perth, Western Australia, Australia

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ABSTRACT

The success of marine management initiatives and our capability for dealing with environmental change largely depend on our understanding regarding the distribution of species and their habitat preferences. In the present study, we deployed baited remote underwater video systems (BRUVs) in a portion of Brazilian central coast (i.e., Todos os Santos Bay) to describe fish species-habitat associations along an estuary-bay-continental shelf gradient. Significant variation in the fish assemblage was found among three ecosystems, four depth classes, and eight different types of habitat, confirming that the structure and composition of fish assemblages is mediated by a set of habitat characteristics forming an ecological mosaic. The highest species richness and relative abundance were found in the mangroves and reefs. The data demonstrated that some species, such as *Lutjanus jocu*, *Lutjanus synagris*, *Carangoides bartholomaei*, *Eucinostomus argenteus* and *Eucinostomus melanopterus* had clear ontogenetic shifts among habitats and across ecosystems. Some species (*Sphoeroides greeley*, *L. synagris*, and *Eucinostomus gula*) were widespread along the ecosystem-level gradient and were observed in a number of different habitats, reflecting more generalist habitat associations. By contrast, a large number of species (54; e.g., *Ptereleotris randali*, *Decapterus macarellus* and *Mugil curema*) were recorded in a single habitat type, indicating they were habitat specialists. Despite this apparent habitat-related pattern, the size-mediated relationships found in many species indicate cross-migration along the ecosystem gradient. Our findings have implications for the conservation and monitoring of fish assemblages highlighting the role of the connectivity of marine habitats as a conservation priority toward to an ecosystem-based management strategy.

1. Introduction

Global trends of coastal development impose pressures on adjacent marine ecosystems, such as estuaries, embayments, gulfs and nearshore regions, and the biodiversity that inhabits these areas (Pittman et al., 2011; Halpern et al., 2015; Portugal et al., 2016). Historically, coastal areas and their biozones (Harvey et al., 2013a) have tended to attract human settlements and the establishment of ports, due to their natural protection from the open ocean and the plentiful resources provided by these highly productive environments (Able, 2005; Weinstein, 2008). Estuarine and coastal ecosystems are particularly impacted by activities such as overfishing and land reclamation which result in habitat loss (Kappel, 2005; Worm et al., 2006; Cloern and Jassby, 2012; Igulu et al., 2014; Halpern et al., 2015).

The development of successful conservation, fisheries management and restoration strategies for aquatic ecosystems depends on a sound scientific understanding of the relationships between the characteristics of the ecosystem and the distribution of fish diversity patterns. These include knowledge about fish-habitat associations, habitat sharing and partitioning, the use of essential habitats during different life phases as well as the importance of habitat connectivity (Lowry and Suthers, 1998; Able, 2005; Bortröm et al., 2011; Pittman et al., 2011; Childs et al., 2015; Hammerschlag et al., 2017). The type of habitat, for example, has proven to be a good predictor of the local fish assemblage, although the more mobile species, in particular, will depend on a variety of habitats and depth zones over the course of their life histories (Heyns-Veale et al., 2016; McLean et al., 2016; Wellington et al., 2018). Given this, focusing conservation and fisheries management strategies and monitoring efforts on only specific critical habitat types may potentially limit the understanding of the dynamics of these systems and the effectiveness of the management (Worm et al., 2006; Fitzpatrick et al., 2012; Heyns-Veale et al., 2016).

* Corresponding author.ICHTUS soluções em meio ambiente, Salvador, BA, Brazil.
E-mail address: amorim_agua@yahoo.com.br (J.A. Reis-Filho).

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The concept of habitat connectivity, for example, provides an important perspective in both terrestrial and aquatic ecology (Olds et al., 2012). One of the gains of conservation priorities is the adoption of strategies to understand the biota-habitat relationships (Beger et al., 2008), for example, the importance of nursery habitats in the perspective of 'seascape nursery', consisting of multiple mosaics of habitat patches that are functionally connected (Nalgerkeken et al., 2015). Recent research in marine ecology has also introduced the depth dimension in their analyses of complex marine-estuarine mosaics, recognizing its importance in tropical seascapes as a component of fish migration and use of refuge habitats (Bradley et al., 2017; Lindfield et al., 2016; Wellington et al., 2018). The distribution and migration patterns of marine fish are determined, typically, by the degree of habitat connectivity, whereas habitat fragmentation will potentially hamper the recovery of fish populations from natural and anthropogenic disturbance (Able, 2005; Olds et al., 2012; Vargas-Fonseca et al., 2016).

Most ecological studies of fish-habitat association and distribution patterns have focused on a limited number of species or habitat types (Bortröm et al., 2011; Olds et al., 2012; Igulu et al., 2014; Yates et al., 2016). This may be a consequence of the challenging task of simultaneously sampling several habitats. Therefore, obtaining precise and accurate information on the diversity, abundance and habitat preferences of fish populations, preferably through remotely operated non-lethal and non-destructive sampling methods, is essential for effective conservation and management. The need to sample structurally complex substrate types, which may damage or destroy capture gear (e.g., gillnets, traps and hook and line) and the prohibitive costs of employing several divers applying UVC technics, have led to the emergence of underwater video sampling methods, such as the Baited Remote Underwater Video (BRUV) system (Harvey et al., 2001; Langlois et al., 2012). This method is known to produce less biased estimates of species richness and relative abundance in comparison with UVC scuba divers or even Diver Operated Video (Harvey et al., 2001, 2002; Langlois et al., 2010; Watson et al., 2010). Despite their ample applicability, only a handful of fish studies have used BRUVs in estuarine ecosystems, and even these have investigated only a very small sample of marine habitats, such as reefs, seagrass beds, and sandy bottoms (Gladstone et al., 2012). This type of research has been particularly rare in the tropics (Bradley et al., 2017), and nonexistent in the southern Atlantic Ocean.

In the present study, we used BRUVs for a multi-habitat approach, focusing on the fish-habitat associations from eight types of marine habitat along an estuary-bay-continental shelf gradient in a nearshore system of the southwestern Atlantic. To evaluate the role of habitat connectivity in the structuring of fish assemblages, we quantified the assemblages in a depth-stratified horizontal ecosystem gradient. Our specific aims were to (i) evaluate the influence of the type of marine habitat, depth zone, and ecosystem (gradient) on the species richness, abundance, and composition of the fish assemblage, and (ii) identify ontogenetic shifts in fish-habitat association patterns.

2. Material and methods

2.1. Study area

BRUV samplings were conducted in Todos os Santos Bay (TSB) in northeastern Brazil (12°50′ S, 38°50′ W; Fig. 1), which is the country's second largest coastal bay with an area of 1223 km^2. The body of water is dominated by bi-directional currents that are stronger during the ebb tide throughout most of the bay (Lessa et al., 2001). The tidal regime is characterized by symmetric diurnal tides ranging between 2.1 and 2.4 m (Cirano and Lessa, 2007). The mean annual rainfall of the study area is ~2400 mm (mean monthly rainfall of ~210 mm) and rainy season typically lasting from March to August (mean monthly precipitation ~280 mm), and a dry season from September to February (~110 mm per month). The embayment is bordered to the east by Salvador, the fourth largest Brazilian city and the state capital of Bahia, with a population of nearly three million inhabitants. The northern margin of the bay includes the site of the Landulpho Alves Refinery, the largest petrochemical complex in the southern hemisphere.

2.2. Defining habitat types

A number of different types of marine habitat were identified *a priori*, based on the pre-existing data set available for the study area (Amado-Filho et al., 2008; Marins et al., 2008; Hatje and Andrade, 2009). Habitat maps were also compiled from a number of different sources, depending on whether they were inshore (i.e., local references) or offshore (i.e., field surveys). These data on the geomorphology and habitat types available in shallow waters (e.g., mangroves, seagrass) and the subtidal zones were used to define the sampling sites for the different inshore/offshore and habitat/depth categories along the estuary-bay-continental shelf ecosystem gradient (Marins et al., 2008; Hatje and Andrade, 2009; Reis-Filho et al., 2014, 2016; Nunes et al., 2015; Reis-Filho et al., 2018a). Based on the available data, eight habitat types were sampled: (1) mangroves, (2) reefs, (3) rocky, (4) muddy and (5) sandy bottoms, (6) seagrass beds, (7) rhodoliths, and (8) soft coral, including *Neospongode* beds, which are substantial areas dominated by colonies of plant and animal organisms that support a rich marine diversity (Fig. 1).

2.3. Experimental design

Four random, replicate BRUV samples were deployed in each of the eight marine habitat types (fixed, 8 levels: sandy, muddy and rocky bottom, reefs, mangroves, seagrass and rhodolith beds, and soft coral) found in the four depth zones (fixed, 4 levels: 1–2 m, 4–10 m, 10–20 m, and 20–30 m) and three ecosystems (estuary, bay, continental shelf). As some habitats are only found at specific depths or are restricted to a given ecosystem, e.g., mangrove, there were some deviations in the study design, resulting in a total of 156 BRUV samples.

2.4. Sampling of fish communities

Baited remote underwater video systems (BRUVs) were deployed temporally and random to collect data on the species richness, abundance and size (body length) of the fish occupying each habitat type, between January and March 2014, corresponding to the dry season in the study area, when the conditions of visibility are at their best (Reis-Filho et al., 2016; Reis-Filho et al., 2018a). The use of BRUVs facilitated the standardization of the sample area during the analysis of the video footage by holding the horizontal range of each sample constant, thus controlling for the potential variation in the visibility among sampling sites (Harvey et al., 2003, 2004). The sampling area was standardized by excluding fish that were located beyond the 5 m horizon of visibility of the BRUVs (a procedure similar to that applied by Fitzpatrick et al., 2012). This allowed us to estimate fish abundance consistently and reliably from all sample of video footage (Watson et al., 2007). The salinity and dissolved oxygen concentrations (mg/l) of the water were measured using an electronic Hanna® depth gauge at the same sites used to obtain the fish samples. We used a Niskin sampler to obtain aliquots of water in depth zones where the BRUVs were deployed. This approach intended slightly assess the water and guarantee that the sampling sites did not present great differences in physical-chemical conditions beyond of already expected estuarine-bay-continental shelf gradient, once it could cause biases in our results.

Single GoPro Hero 3 + Black Edition digital high-definition cameras (www.gopro.com) with a wide-angle lens and fixed focal length were used to obtain the video footage, at 30 frames per second. The cameras had a resolution of 1920 × 1080 pixels. These cameras are a cost-effective, yet reliable alternative to the handheld cameras

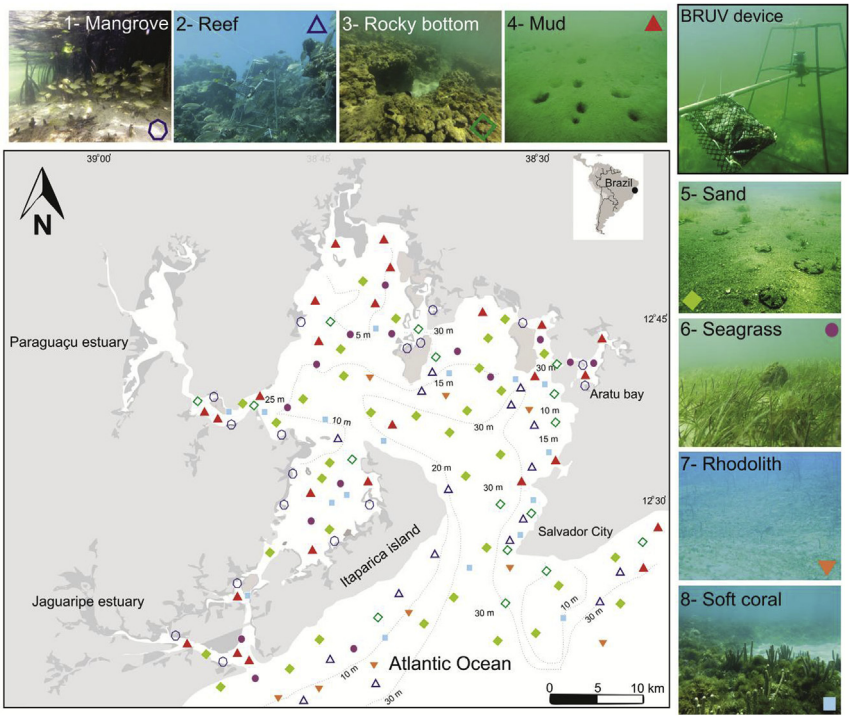

Fig. 1. Study area detailing BRUV sampling locations for the depth-habitat combinations in each of the three ecosystems (seascapes) of the Todos os Santos Bay, Brazil.

commonly used in BRUV studies (Struthers et al., 2015; Schmid et al., 2017). The cameras were powered by lithium batteries (1180 mAh), allowing for a maximum duration of approximately 130 min of recording. The entire camera system was protected in a watertight case, mounted on a galvanized iron support (Cappo et al., 2003, 2004). The bait bags were filled with approximately 800 g of crushed sardine (*Sardinella* spp.) for each sample. To avoid bias from overlapping bait plumes, all BRUVs were deployed at a minimum distance of at least 500 m from one another. The BRUVs were retrieved after recording for 70 min at each location, and then prepared and rebaited for deployment at the next site.

2.5. Image analysis

Samples were analyzed by one person only using Adobe Premier v.6.0. All fish were identified to the lowest possible taxonomic level. The Maximum Number (MaxN) of individuals of a given species present at one time within the camera's field of view was determined (Willis and Babcock, 2000; Cappo et al., 2003, 2004; Harvey et al., 2007; Gladstone et al., 2012; Harvey et al., 2013b). The MaxN is a conservative measure used to avoid the repeated counting of individuals, but tends to underestimate abundance, in particular in the case of schooling fish (Schobernd et al., 2014; Campbell et al., 2015). Body length estimates were obtained by comparing the fish with the bait bag (approximately 20 cm) when equidistant from the camera. The body size could be estimated more accurately when the fish would often bite the bait bag from different angles. To ensure the accuracy of these estimates, it was necessary for the entire fish to be visible in the frame, a procedure similar to that applied by Harvey et al. (2002). Known age-related variation in coloration patterns, which are typical of some species, were also taken into consideration.

2.6. Data analysis

2.6.1. Univariate tests

The species richness (S) and estimates of relative abundance (MaxN) were analyzed using standard descriptive and comparative (factors) statistical procedures. Following our experimental design, the variation in these two parameters (S and MaxN) was analyzed among ecosystems, depth, and habitat type using a univariate permutational analysis of variance (PERMANOVA) based on Euclidean distance matrices with 9999 permutations (Anderson, 2001). *Post hoc* pairwise comparisons were evaluated using the PERMANOVA *t* statistic, with a $P \leq 0.05$ significance level.

2.6.2. Multivariate tests

As our sampling design was explicitly unbalanced, since mangrove and other shallow habitats do not occur in deeper zones and deeper habitats (e.g., rhodoliths or soft corals) either do not occur or are sparse in the shallower habitats, a permutational multivariate analysis was used to test for variation in the composition of the fish assemblage among the different ecosystems, depth zones, and marine habitat types (PERMANOVA; Anderson, 2001; Anderson et al., 2008). The statistical significance of this analysis was tested using 9999 permutations of the residuals in a reduced model and Type III (partial) sums of squares (SS; Anderson et al., 2008). The PERMANOVA were run using a Bray-Curtis similarity matrix calculated for the fourth root transformed biological data.

Significant ($p < 0.05$) differences between groups were investigated further using a canonical analysis of principal coordinates (CAP; Anderson and Willis, 2003), a constrained ordination technique that produces predictive models maximizing the differences among *a priori* groups and highlighting patterns that may be cryptic in unconstrained ordinations. The ability of the model to discriminate an *a priori* group was assessed by a "leave-one-out" allocation and cross-validation test (Lachenbruch and Mickey, 1968; Anderson et al., 2008). A Spearman correlation of at least 0.7 was used as an arbitrary limit for the inclusion of potential correlations between fish abundance and categories relative to the canonical axes (Anderson et al., 2008).

To assess the distribution of species among the marine habitats, a shade plot was produced by clustering the samples on the x-axis based on Bray–Curtis similarity, with the 30 most common and abundant

species (based on Whittaker's index of association: Clarke and Gorley, 2015) on the y-axis. The similarity profile test (SIMPROF) was used to determine the significance of the differences between the clusters and to identify groups of fish species that present a degree of habitat partitioning (Clarke and Gorley, 2001). The habitat-related abundance of each species was used to interpret its role as a habitat generalist or specialist (Appendix 1). All the statistical analyses were run in PRIMER version 7.0.11 with the PERMANOVA + add-on (Anderson et al., 2008; Clarke and Gorley, 2015).

2.6.3. Distribution of body length frequencies

To verify the ontogenetic variation in the fish-habitat associations of a selection of species (including the most abundant and/or commercially important taxa), Kernel Density Estimates (KDEs) were run using the body length frequency data from each habitat type. This analysis was run in the R software (R-Team, 2014) (Langlois et al., 2012).

3. Results

3.1. Environmental characteristics

The mean salinity (± S.D.) in the study area reflected the gradient from the estuary (27.02 ± 0.51), passing through the bay (33.13 ± 1.36) to the continental shelf (36.39 ± 0.78). Mean dissolved oxygen levels (mg/l) were stable along this horizontal gradient, however, with only slight variation between the bay (8.21 ± 0.52) and continental shelf (8.34 ± 0.29). At a habitat level, the mean salinity and dissolved oxygen levels were consistent with the seascapes described above.

3.2. Descriptive statistics and univariate PERMANOVA

We recorded a total of 3989 individual fish from 119 taxa belonging to 48 families. It was possible to identify 109 taxa to species, while the remaining 10 were identified only to the genus level. The most diverse families were the Carangidae, Haemulidae, Serranidae and Gerreidae, with 12, 11, 9 and 7 taxa being recorded during the study, respectively. Overall, the most abundant species, based on the total (TMaxN) and mean relative abundance (mean MaxN ± SD) per sample (Appendix 1) were *Haemulon aurolineatum* with a total of 469 individuals (3.01 ± 7.58), *Eucinostomus argenteus* with 437 individuals (2.80 ± 3.84), *Eucinostomus melanopterus* with 232 individuals (1.49 ± 6.24), *Eucinostomus gula* with 182 individuals (1.17 ± 2.27), *Diplectrum radiale* (1.01 ± 1.57) and *Lutjanus synagris* (1.01 ± 1.55), both with 157 individuals each.

Considering our complete dataset, almost half (n = 54) of the 119 fish taxa presented very specialized patterns of habitat use, being recorded in only one of the eight habitat types sampled. No species was ubiquitous, although five taxa occurred in seven habitats, and 25 taxa were recorded in half (4) of the habitat categories (Appendix 1). These habitat generalists were predominantly mojarras and jacks, with the exception of *Sphoeroides greeleyi* (Tetraodontidae) and L. *synagris* (Lutjanidade). The largest numbers of species found exclusively in a single habitat type were 14 in muddy habitats and 13 in reefs, followed by sandy habitats, with 12 exclusive species.

No significant variation was found among the bay, estuary and continental shelf ecosystems in total relative abundance of fish (TMaxN) (univariate PERMANOVA: $p = 0.1624$) or species richness (S) ($p = 0.2614$) (Fig. 2-A), although highly significant differences were found in both variables among different depths (MaxN: $p < 0.0001$; S: $p = 0.0007$) and habitat types (MaxN and S: $p < 0.0001$) (Tables 1 and 2). The highest relative abundance (mean MaxN ± Standard Error: 37.8 ± 5.06) and species richness (mean S ± SE: 7.4 ± 0.63) were recorded in the shallowest depth zone (1–2 m), whereas the lowest abundances were associated to 20–30 m (MaxN: 18.9 ± 3.77; S: 5.6 ± 0.71) (Fig. 2-B) as confirmed by the pairwise PERMANOVAs

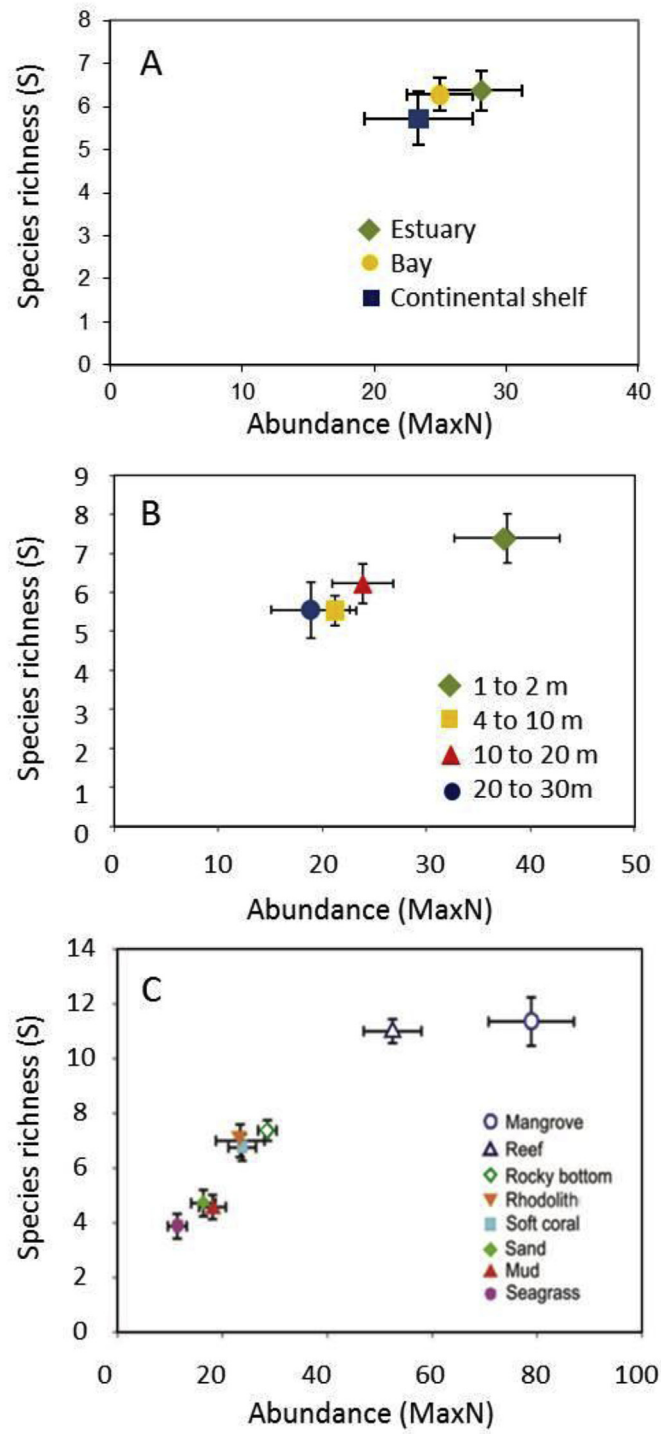

Fig. 2. Mean fish species richness (S) and relative abundance (MaxN) recorded in the different ecosystems, depths, and marine habitat types sampled with BRUV in Todos os Santos Bay, Bahia, Brazil.

(Appendix 2). On average, the BRUVs deployed in the mangrove habitats scored the highest relative abundance (78.9 ± 8.15), followed by reef habitats with 52.4 ± 5.50. In all other habitats, the mean MaxN values were at least 50% lower (Fig. 2-C). Species richness was also the highest in the mangrove (11.4 ± 0.89) and reef habitats (11.0 ± 0.44), followed by rocky habitats (7.4 ± 0.38) and rhodolith (7.0 ± 0.60) (Fig. 2-C). The lowest relative abundance (11.4 ± 1.80) and species richness (3.9 ± 0.46) were recorded in seagrasses. Most pairwise PERMANOVAs indicated that mangroves, reefs and rocky

Table 1
Univariate PERMANOVA Result for Relative Abundance (MaxN) for the tested factors ecosystem (Ec), depth (De), marine habitat (Ha) and the interaction (x) between Ec and De, Ec and Ha and De and Ha.

Source	df	SS	MS	Pseudo-F	P(perm)	Unique perms
Ec	2	514.72	257.36	1.8507	0.1624	9945
De	3	7033.2	2344.4	16.859	**0.0001**	9954
Ha	7	40746	5820.8	41.858	**0.0001**	9935
EcxDe	4	1557.4	389.35	2.7999	**0.0298**	9952
EcxHa	7	4129.7	589.96	4.2424	**0.0003**	9933
DexHa	10	4441.3	444.13	3.1937	**0.0029**	9934
EcxDexHa	5	422.05	84.409	0.60699	0.69	9946
Res	117	16270	139.06			
Total	155	75114				

Table 2
Univariate PERMANOVA Result for Species Richness (S) for the tested factors ecosystem (Ec), depth (De), marine habitat (Ha) and the interaction (x) between Ec and Ha and De and Ha.

Source	df	SS	MS	Pseudo-F	P(perm)	Unique perms
Ec	2	10.058	5.0288	1.3297	0.2614	9888
De	3	73.01	24.337	6.4347	**0.0007**	9954
Ha	7	938.71	134.1	35.457	**0.0001**	9959
EcxDe	4	22.685	5.6713	1.4995	0.2121	9949
EcxHa	7	134.69	19.242	5.0877	**0.0001**	9942
DexHa	10	102.96	10.296	2.7223	**0.0051**	9927
EcxDexHa	5	4.3628	0.87256	0.23071	0.9442	9936
Res	117	442.5	3.7821			
Total	155	1729				

bottoms accounted for the main variations in abundance and species richness. (Appendix 3).

3.2.1. Multivariate analysis

The overall multivariate PERMANOVA revealed significant differences in the composition of the fish assemblages ($p < 0.0001$ in all cases) among the three ecosystem types, the four depth levels, and the marine habitats (Table 3). Pairwise comparisons across the depth levels revealed significant differences in the fish assemblage composition between the shallow (1–2 m) and all other depths (< 0.0001), yet not among the other depth levels (Appendix 2), whilst across the marine habitats, all fish assemblage compositions differed significantly in multivariate pairwise comparisons (most with $P < 0.0001$) (Appendix 3). There were significant interactions for ecosystem vs. habitat and depth vs. habitat (Table 3) with more diversified fish fauna assigned to mangrove and reef habitats located in shallow water inner to bay (Fig. 2).

We used a Canonical Analysis of Principal Coordinates, CAP (Table 4) to test our *a priori* hypothesis on the factor "habitat" evidenced in multivariate pairwise PERMANOVAs as structuring factor (Fig. 3). The average cross-validation ranged from 53% to 100% (Table 4), and all (100%) the samples from mangrove and rhodolith habitats were classified correctly, followed by reefs, at 81%, and muddy habitats, at 71%. According to CAP, habitat arrangement in the multivariate space clearly revealed a separation along the CAP1 axis, into consolidated (black arrow) and unconsolidated habitats (red arrow) (Fig. 3). Moreover, the CAP ordination indicated that the rhodolith beds host a distinct assemblage of marine fishes, forming an isolated and non-overlapping group (Fig. 3). A similar, but less exclusive grouping was formed by the reef habitat samples. The fish assemblages recorded over soft coral and rocky bottom habitats were also relatively well-defined, with reduced overlapping. By contrast, the fish assemblages recorded in muddy, sandy, and seagrass/macroalgae habitats, over unconsolidated substrates, formed a single, highly overlapping group, reflecting their broad similarities (Fig. 3). The mangrove fish assemblage formed a small concentrated group, with similarities to both consolidated and unconsolidated categories (Fig. 3).

Table 3
Multivariate PERMANOVA results for the composition of the fish assemblages according to ecosystem (Ec), depth (De), marine habitat (Ha), and the interactions (x) among the three factors.

Source	Df	SS	MS	Pseudo-F	P(perm)	Unique perms
Ec	2	48328	24164	11.752	**0.0001**	9914
De	3	29854	9951.4	4.8399	**0.0001**	9876
Ha	7	1.44E+05	20609	10.023	**0.0001**	9838
EcxDe	4	9602.1	2400.5	1.1675	0.2134	9859
EcxHa	7	3.34E+04	4765.1	2.3175	**0.0001**	9835
DexHa	10	3.50E+04	3496.2	1.7004	**0.0001**	9793
EcxDexHa	5	8188	1637.6	0.79645	0.8697	9845
Res	108	2.22E+05	2056.1			
Total	146	5.31E+05				

The shade plot that combines the two cluster analyses of the fish assemblage data identified four separate groups of marine habitat types (top dendrogram) and three principal groups (left dendrogram) of fish taxa with similar habitat associations (Fig. 4). Mangrove (Fig. 4-A) and Rhodolith (Fig. 4-D) habitats each formed an isolated profile. A third group (Fig. 4-B) was formed by the habitats with unconsolidated substrates (sand, mud, and seagrass/macroalgae), while the fourth group (Fig. 4-C) included the habitats with consolidated substrates (reefs, rocky bottoms, and soft coral). Cross-referencing these clusters with the 30 most abundant and frequent fish species recorded during the study, a group (Fig. 4-I) of 11 species from a number of different families was associated strongly with mangrove habitats. This group included *Mugil curema*, which was recorded exclusively in the mangrove habitat, *Lutjanus alexandrei*, recorded mainly in the mangrove, but also in reefs, and *Eucinostomus melanopterus*, which was recorded primarily in mangrove, but also in soft coral habitats. A second group (Fig. 4-II) of 12 species, was detected in a range of habitats from the two groups (consolidated and unconsolidated substrates). These species were typically more abundant and frequent over reef, rock and soft coral than seagrass, sand or mud. The third group (Fig. 4-III) contained seven species related predominantly with hard bottom habitats, mainly with rhodolith, followed by reefs. *Oligoplites* sp., somewhat of an exception within this group, was also detected in most other habitats, in particular rhodoliths.

Different species, even within the same family, presented varying degrees of habitat partitioning (Fig. 4). In the case of the family Lutjanidae, for example, *Lutjanus alexandrei* and *Lutjanus jocu* were most often associated with mangrove habitats, while *Ocyurus chrysurus* was associated predominantly with reefs, and *Lutjanus synagris* was found in all habitats except the mangrove, being associated primarily with rhodolith and seagrass. By contrast *Oligoplites* sp., *Ptereleotris randalli* and *Decapterus macarelus* were most common in rhodolith habitats.

3.2.2. Kernel Density Estimates of length classes

Kernel Density Estimates of the length classes revealed distinct, ontogenetically-driven habitat occupation patterns for the six most abundant fish species (Fig. 5). In the family Lutjanidae, the smallest individuals of *L. jocu* were recorded exclusively in mangrove habitats, whereas medium to large sized fish (20–40 cm) were recorded mainly over rocky bottoms and, to a lesser extent, soft coral and rhodolith, and the largest specimens (> 40 cm) occurred only in reef habitats. This reflects a clear ontogenetic shift from estuarine habitats (mangrove) to the bay and continental platform (reef) with increasing body size (Fig. 5–A). Similarly, while the smallest individuals (5–10 cm) of *L. synagris* were recorded exclusively in seagrass, medium-sized specimens (10–20 cm) were more frequent over sandy bottoms and rhodolith, and larger individuals (20–40 cm) were found mainly over muddy and reef habitats (Fig. 5–B).

In the case of the family Carangidae, the probability of recording

Table 4
The Canonical Analysis of Principal Coordinates (CAP) leave-one-out allocation of habitat types based on fish abundance composition obtained by BRUVs in Todos os Santos Bay, Bahia, Brazil.

Original group	Mangrove	Mud	Soft coral	Sand	Seagrass	Reef	Rocky bottom	Rhodolith	% correct
Mangrove	8	0	0	0	0	0	0	0	100
Mud	1	22	0	6	1	0	1	0	71
Soft coral	0	0	9	0	0	3	4	0	56
Sand	0	5	1	23	7	0	0	1	62
Seagrass	0	0	3	3	8	1	0	0	53
Reef	0	0	0	0	0	13	2	1	81
Rocky bottom	0	0	3	0	0	1	12	0	75
Rhodolith	0	0	0	0	0	0	0	8	100
	(m = 15)								

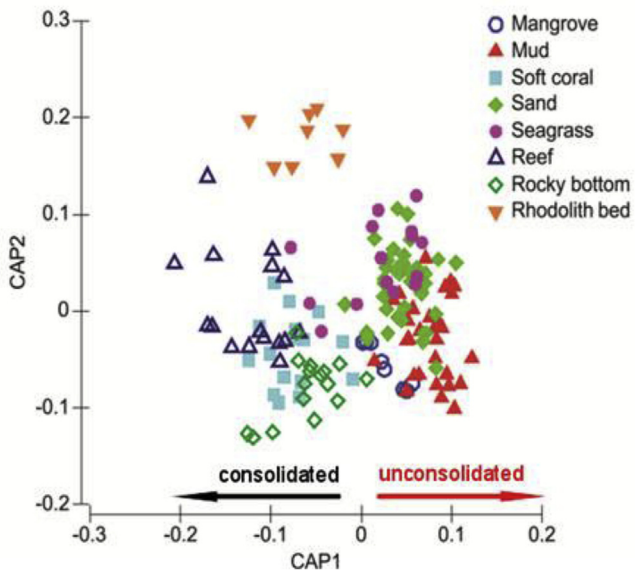

Fig. 3. Canonical Analysis of Principal Coordinates (CAP) of the composition of the fish assemblages sampled by BRUVs in the different marine habitat types of Todos os Santos Bay, Bahia, Brazil.

small (< 20 cm) *Caranx bartholomaei* was highest over sand and in mangrove habitats, while medium-large fish (< 30 cm) were most frequent in seagrass and reefs, although some of the largest specimens were also recorded over soft coral (Fig. 5–C). In the haemulid *Haemulon aurolineatum*, the smaller fish (< 15 cm) were recorded only in seagrass habitats, while specimens larger than 15 cm were found in all habitats with consolidated substrates, and the largest (> 20 cm) individuals were found exclusively on reefs (Fig. 5–D). In *Eucinostomus gula* (Gerreidae), the smallest individuals (< 10 cm) were recorded exclusively in seagrass habitats. Larger fish (> 10 cm) were ubiquitous in the habitats with unconsolidated substrates, in particular mud, sand, and seagrass, but also over soft coral and rocky bottoms (Fig. 5–E). Similarly, the smaller specimens of *Eucinostomus argenteus* were more frequent over mud, sand and seagrass, whereas larger individuals (> 10 cm) were more abundant over muddy and sandy bottoms, but also over soft coral and rocky bottoms (Fig. 5–F).

4. Discussion

Establishing baselines for the definition and understanding of fish-habitat associations is an essential component to ecosystem approach in fishery management, biodiversity conservation and the spatial planning of coastal development (Browman and Stergiu, 2004; Harvey et al., 2013a). The use of baited remote underwater video systems (BRUVs) enabled us to characterize the organization of the wealth fish biodiversity of the Todos os Santos Bay' marine ecosystems, based on the occupation of distinct habitats by well-defined fish assemblages. Our study is the first meso-scale investigation of fish communities using a single standardized sampling technique for a variety of habitat types along an estuarine-bay-continental shelf gradient in the South Atlantic Ocean. In addition to identifying more than half of the 119 registered taxa as habitat specialists and showing multi-habitat associations for generalist species, the results highlighted ontogenetic variations in the distribution of certain fish species, as well as the role of key habitats during their life history.

Yet, estuaries and bays are hydrologically very dynamic environments, and such conditions may affect the BRUV-derived data to a certain degree, due to potential variation in the bait plume dispersion (see Taylor et al., 2013). Bait plume behavior has rarely been modeled (Priede and Merrett, 1996) and is still a key issue in using BRUVs, while a couple studies compared baited and unbaited video systems (Watson et al., 2005; Harvey et al., 2007), also to evaluate the selectiveness of bait attraction. In our study we intended to minimize such bait plume biases by carefully deploying the BRUVs in areas with lower current speed and whenever possible during the slack tide. Thus way, we seek to ensure that bait disperses into the water column to form a plume of attractant down-current of the BRUV, acting as an olfactory stimulus to attract fish located not too far, which could create a bias due to bait plume area variation and produce an associated effect on relative sampling effort.

As expected, the differences found in the fish assemblages along the longitudinal ecosystem gradient were probably related primarily to habitat characteristics (Cappo et al., 2011), in which range in temperature and salinity were not a potentially confounding nuisance but understood as components of their habitats. While the univariate analysis (S and MaxN) for the ecosystem factor (Tables 1 and 2) did not reveal statistically significant differences, mean species richness was higher in the estuary and bay ecosystems in comparison to continental shelf. The embayment zone represents a transition between inland waters (i.e., estuaries) and coastal areas exposed to oceanic forces, which is occupied by a diverse fish assemblage, as observed in similar regions around the world (Ray, 2005; Sheaves, 2006). The accentuated species diversity found in the inner Todos os Santos Bay (TSB) is probably an ecological response to its geomorphological complexity, which accommodates a large variety of marine habitats. In tropical regions, the morphology of coastal areas is recognized as playing a fundamental role in the structuring of coastal fish abundance and high species diversity patterns (Sheaves, 2006).

Depth is an important predictor of marine fish assemblages and recent studies have confirmed its structuring role (Drazen, 2007; Zintzen et al., 2012; Heyns-Veale et al., 2016; Sih et al., 2017; Wellington et al., 2018) as a result of adaption to the physically challenging environment (high water column pressure, low temperature and light and other factors) (Zintzen et al., 2012). Despite the range covered in the TSB was reduced to a maximum depth of 30 m, the distribution of the taxa did not simply result in a homogenized < 30 m

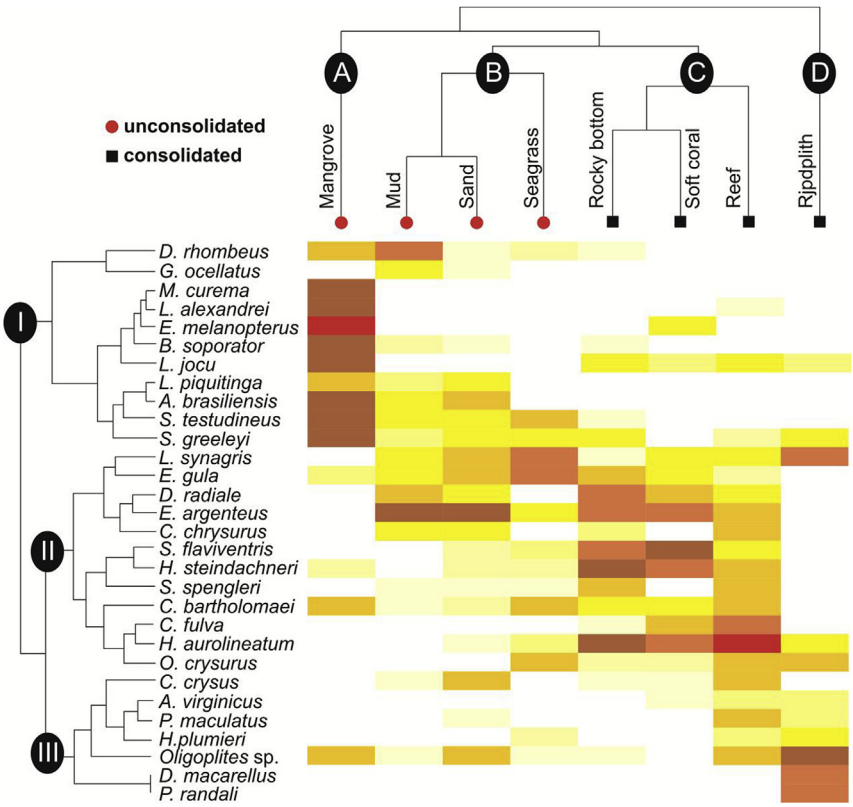

Fig. 4. Shade plot of the distribution of fish species among the four groups of marine habitat types (top dendrogram), showing the three groups (left cluster) of the 30 most frequent and abundant fish species and their habitat associations. The colour gradient from black (strong correlation) to soft green (weak correlation) indicates the strength of species-habitat association. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

shallow-water assemblage, but revealed differentiated fish assemblages along this depth gradient, notably between the shallow (1–2 m) and the deeper (10–20 m and 20–30 m) depth zones (Appendix 2). This may indicate that even in restricted depth variations the fish fauna can be organized due to both, previously missing components (i.e., depth variation) and sharp differences in fauna-habitat relationships, for example can be determined by depth-associated habitats and shallow zones with strong tidal influence (Bradley et al., 2017; Reis-Filho et al., 2016).

Despite the fact that transitional waters (i.e., estuaries, bays and whole coastal zone) act as mosaics of interconnected habitats (Sheaves, 2009), the conditional and transient exposition of shallow zones to intertidal effects (i.e., tidal rhythms whereby the microhabitats expand and shrink according to the rise and fall of water level) resulted in distinct fish assemblages, varying in their composition, with the low-depth zones accounting for greater abundance and richness. While abundant generalist species in our study contributed to a more homogenized fish assemblage along the analyzed ecosystem gradient, mainly in subtidal zone (see sand and mud habitats in Shade plot – Fig. 4), mangrove, reef and rhodolith beds presented unique fish assemblage with greater abundance of habitat-specialist species. The strong interactions between the factors depth and habitat, detected in both PERMANOVA univariate and multivariate analyses, indicated that the depth-related differences in the fish fauna were mainly influenced by specific habitats (Tables 1–3). As such, mangrove – exclusively a low-depth habitat – contributed to these habitat-driven depth patterns with the highest species richness and abundances per habitat (Fig. 2). Apart from the 1–2 m zone, both mean S and MaxN varied much less amongst the other three depth zones. A prominent characteristic of fish assemblages in shallow waters is that there are several resident species (Barletta et al., 2003; Reis-Filho et al., 2011) that do not move over large distances during their tidal migration (Horn et al., 1999) as well as the potential addition of new species coming from tidal movements (Reis-Filho et al., 2016). This further highlights the ecological importance of shallow habitats in the intertidal zone of coastal areas, with special emphasis on mangroves, by offering abundant resources and protection from predation for a wide range of marine-estuarine fish (Igulu et al., 2014).

Additionally, we also argue that the fishing pressure in the TSB is widespread and not limited to the shallow depths (Soares et al., 2011; Reis-Filho et al. 2014, 2018a; Reis-Filho, 2016; Reis-Filho et al., 2016b) acting as a critical driver, once local fisheries typically target large-bodied species, mainly the piscivorous-carnivorous fish, as evidenced in the intensity picture of fishing pressure contained in Reis-Filho et al. (2018a). Thus, the relative abundances in this trophic group can provide a useful indicator for the existing fishing pressure (Weijerman et al., 2013). In this sense, the lack or lesser extent of records of some large or even top predators in all four depth zones, such as sharks and large groupers (e.g. *Mycteroperca* spp.) (Appendix 1) and confirmed by overlap of our BRUV site deployments with areas of high fishing pressure, further reinforce that the processes driving the estuary-bay-continental shelf trophic interactions may be affected by the intensive fishing activity in all habitat types and depth zones.

In the present study, strong links were also identified between the size-related fish assemblages and the distribution of habitats along the estuary-bay-continental shelf gradient. As in other marine embayments around the world (e.g., Wakefield, 2010; Wakefield et al., 2011, 2013; Berkström et al., 2013), the TSB likely provides important nursery habitats for fish (Reis-Filho et al., 2016; Reis-Filho et al., 2016) and supports high fishery harvests (Hatje and Andrade, 2009; Soares et al., 2011; Reis-Filho, 2016; Reis-Filho et al., 2018b). The patches of mangrove and seagrass provide primary habitats for the juvenile fish (see Reis-Filho and Leduc, 2018) that presented size-structured (ontogenetic) cross-habitat shifts, while the largest individuals were registered most frequently over consolidated substrates, notably reef and rocky bottom habitats (Fig. 5). Together with local fishery data, which confirm the abundance of large individuals of commercially targeted species on the continental shelf, off the mouth of the TSB (Nunes et al., 2012; Reis-Filho, 2016), our findings support the conclusion that, in this coastal region, many fishes such as snappers (*Lutjanus synagris* and

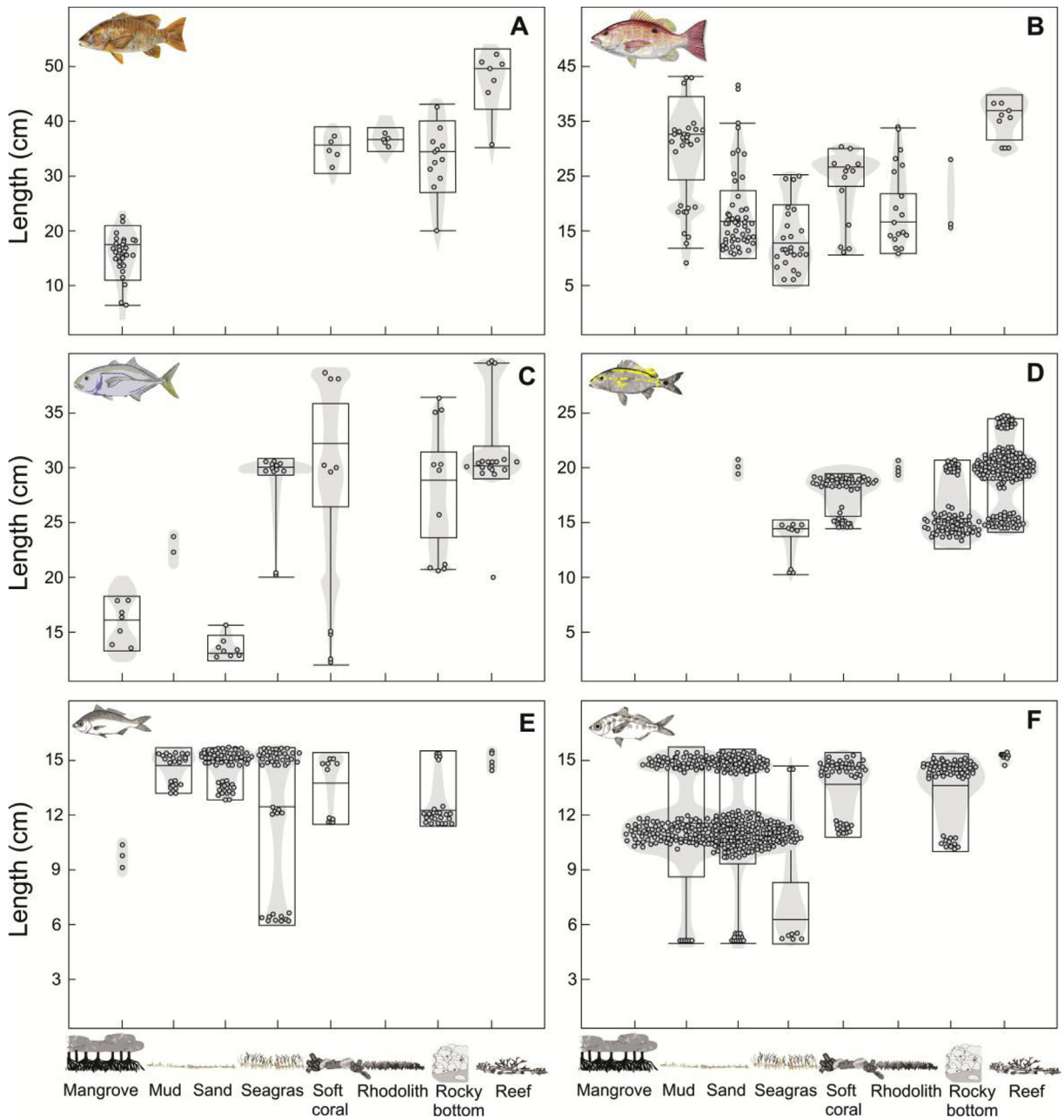

Fig. 5. Kernel Density estimates for the different length classes of six abundant fish species with multi-habitat associations in Todos os Santos Bay, Bahia, Brazil. The upper and lower limits of the box indicate the 1st and 3rd quantiles, the inner line of the box indicates the average on the data distribution and larger widening of the shading area indicates greater probability of occurrence of the observed pattern. A: *Lutjanus jocu*; B: *Lujanus synagris*; C: *Caranx bartholomaei*; D: *Haemulon aurolineatum*; E: *Eucinostomus gula*; F: *Eucinostomus argenteus*.

L. jocu) and jacks (*Carangoides bartholomaei*), undergo ontogenetic cross-habitat migrations, from the inner to the outer estuarine habitats of the TSB, with increasing body size. We suggest that ontogenetic shift-based connectivity investigation must be incorporated into monitoring programs, once the complex relationships among multiple elements of space (i.e., habitat types) and fish populations are essential to guarantee the effective conservation and management of TSB.

The cross-movements of fishes within the marine landscape and fish-habitat associations in general are largely unknown in the South Atlantic Ocean (but see Moura et al., 2011), which jeopardizes the identification and protection of the critical pathways of settlement, nursery habitats, and migration corridors threatened by overfishing and escalating human development (e.g., construction of shipping terminals and dredging). Our simultaneous video sampling of fish in distinct marine habitats provided a useful baseline of the abundance and distribution of fish and fish-habitat associations along the tropical estuarine-bay-continental shelf gradient of the TSB. However, whether the connectivity among multiple habitats is important for individual fishes in order to provide a source population for the recolonization of specific habitats, such as mesophotic reefs (see Slattery et al., 2011), or they simply depend on cross-habitat movements to complete their life cycle, remains unclear. BRUV monitoring on a continuous temporal scale, studies of foraging behavior at a meso-spatial scale (Papastamatiou et al., 2015) and analysis of fish movements between tidal and subtidal habitats (Ellis and Bell, 2008; Krumme et al., 2014; Reis-Filho et al., 2016) should go hand in hand with the continuous endeavor of applying principles of ecological connectivity to marine environments.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.marenvres.2019.05.004.

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